

ROBERT VOKÁČ, BSc

Software Engineer – C++ Systems & Cross-Platform Development

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PROFILE

Software Engineer with substantial independent work in **modern C++ systems programming**.

I build **cross-platform frameworks**, compatibility layers, runtime systems, and platform abstractions using C++23, **SDL3**, **OpenGL**, **bgfx**, and **CMake**. My work focuses on portable architecture, API compatibility, rendering/input/audio systems, and application portability across Windows, Linux, WebAssembly, and Android.

Seeking roles in systems programming, framework engineering, tooling, platform abstraction, or cross-platform software development.

RELEVANT C++ ENGINEERING EXPERIENCE

Independent C++ Systems Developer – OpenEggbert

Open-source engineering work

2024 – present

Designed and built modern C++ framework and compatibility-layer projects focused on portability, API behavior reproduction, and platform abstraction.

Main technologies: **C++20/23**, **SDL3**, **OpenGL**, **bgfx**, **CMake**, **Windows**, **Linux**, **WebAssembly**, **Android**

CNA - Cross-Platform C++ Framework

GitHub: <https://github.com/openeggbert/cna> Web: <https://libcna.com>

12k+ LOC cross-platform C++ framework implementing an XNA-style programming model over SDL3, validated by a working C# XNA game port running natively on desktop, Android, and WebAssembly.

- Designed a cross-platform C++ framework layer over SDL3
- Built application loop, input, rendering, audio, and resource systems
- Added rendering backends for **SDL_Renderer**, **OpenGL**, and **bgfx**
- Ported a C# / XNA application to native C++
- Replaced .NET, MonoGame, and original XNA dependencies with native C++/SDL3 code
- Produced stable builds from one codebase for Windows, Linux, WebAssembly, and Android

FreeDirect + FreeAPI - Legacy Graphics & System Compatibility Layer

GitHub: <https://github.com/openeggbert/free-direct>

GitHub: <https://github.com/openeggbert/free-api>

7k+ LOC modern C++ compatibility layer focused on legacy graphics and system APIs.

- Created DirectDraw-style 2D rendering abstractions
- Built event loop, window/input handling, and platform integration
- Enabled legacy Windows applications to run through a modern SDL3-based backend
- Reproduced selected DirectX 3 / DirectDraw and WinAPI-style behavior without obsolete SDK dependencies

- Continued work on broader compatibility for legacy Windows applications

Nova3D – Experimental 3D Runtime Layer

GitHub: <https://github.com/openegbert/nova-3d> **Web:** <https://libnova3d.com>

Experimental 3D runtime layer built on top of CNA.

- Designed an Urho3D-inspired API over CNA
- Implemented scene, camera, mesh rendering, and demo infrastructure
- Explores evolution from runtime compatibility toward 3D engine architecture

Hive - Modular Backend Platform

GitHub: <https://github.com/robertvokac/hive>

44k+ LOC metadata-driven backend platform demonstrating large-scale C++ application architecture, extensibility, and backend systems design.

- Designed a plugin-based architecture with dependency-aware startup and migrations
- Built a custom model/ORM layer with metadata-driven entity definitions
- Implemented auto-generated REST APIs from shared model metadata
- Developed scheduler, trigger, validation, and automation pipelines
- Added authentication, token management, operational logging, and migration integrity mechanisms
- Created schema-driven frontend generation backed by shared backend metadata

PRODUCTION SOFTWARE EXPERIENCE

Software Engineer - Assist

2017 – present

Production software engineering role focused on backend systems, SQL, legacy codebases, and long-term maintenance.

- Maintained and improved production Java applications and SQL databases
- Diagnosed and resolved production incidents in large legacy applications
- Performed root-cause analysis, debugging, and long-term stability improvements

Technologies: Java, SQL, Git, Bash

CORE SKILLS

Languages: C++20/23, SQL, Java

C++ / Systems: STL, RAII, smart pointers, resource ownership, modular architecture

Cross-platform: Windows, Linux, WebAssembly, Android

Graphics / Platform: SDL3, OpenGL, bgfx, DirectDraw-style rendering, platform abstraction

Build / Tools: CMake, Git, Bash, CLion, Visual Studio

Backend / Data: SQL, SQLite, REST API design, ORM concepts

Core areas: compatibility layers, legacy behavior reproduction, rendering/input/audio systems

EDUCATION

BSc in Computer Science

Czech University of Life Sciences Prague

2014 – 2017

LANGUAGES

Czech: Native

English: B2